Anonymity and Tor

CSE 405 - Lecture 13

Outline

- Anonymity
- Proxies and VPNs
- Tor
  - Weaknesses: Timing attacks
  - Weaknesses: Collusion
  - Weaknesses: Distinguishable traffic
- Tor in Practice

Anonymity

Anonymity

- **Anonymity**: Concealing your identity
    - Anonymous communication on the Internet: The identity of the source and/or destination are concealed
- Anonymity is not confidentiality
    - Confidentiality hides the contents of the communication
    - Anonymity hides the identities of who is communicating with whom

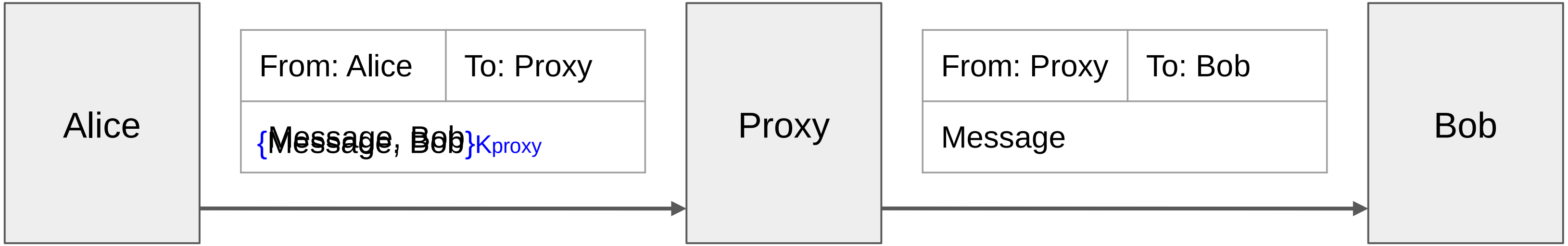
Anonymity on the Internet

- Anonymity on the Internet is hard
  - Difficult, if not impossible, to achieve on your own
  - Packets contain the source IP address and destination IP address
- Anonymity is easier for attackers
  - An attacker can hack into someone else's computer and send communications from that computer
  - We assume honest users won't hack into other computers
- Main strategy for anonymity: Ask someone else to send messages for you

Proxies and VPNs

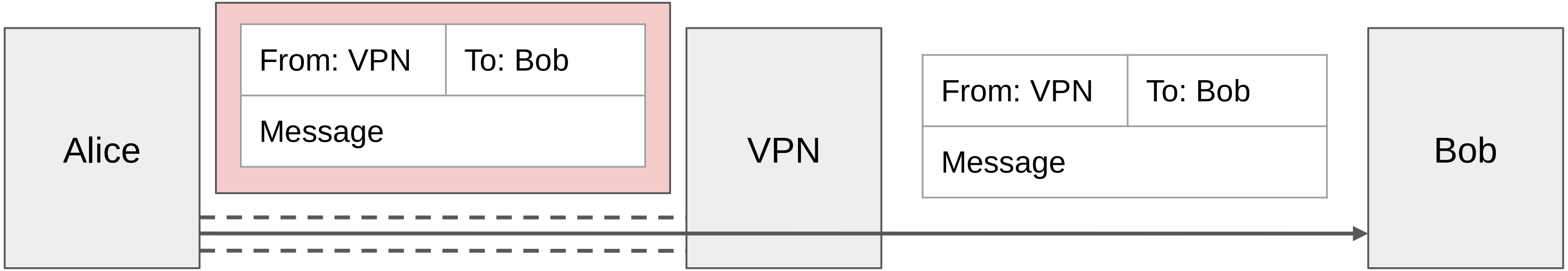
Proxies

- Alice wants to send a message to Bob
  - Bob shouldn’t know the message is from Alice
  - An eavesdropper (Eve) cannot deduce that Alice is talking to Bob
- **Proxy**: A third party that relays our Internet traffic
  - Alice sends the message and the recipient (Bob) to the proxy, and the proxy forwards the message to Bob
    - The recipient’s name (and optionally the message) is encrypted, so an eavesdropper does not see a packet with both Alice and Bob’s identities in plaintext
  - Bob receives the message from the proxy, with no indication it came from Alice



Virtual Private Networks (VPNs)

- VPNs: A virtual connection to an internal network
  - Allows access to an internal network through an encrypted tunnel
  - Creates an alternative use case: Appear as though you are coming from the virtually connected network instead of your real network!
    - Similar concept to proxies, but Alice directly sends packets as though coming from the VPN, wrapped in the VPN's layer of encryption
    - **Proxies operate at the application layer, while VPNs operate at the network layer**



Proxies and VPNs: Issues

- Performance
  - Sending a packet requires additional hops across the network
- Cost
  - VPNs can cost $80 to $200 per year
- Trusting the proxy
  - The proxy can see the sender and recipient's identities
  - Attackers might convince the proxy to tell them about your identity

Tor

Tor

- Idea: Send the packet through multiple proxies instead of just one proxy
- **Tor**: A network that uses multiple proxies (relays) to enable anonymous communications
  - Stands for **T**he **O**nion **R**outer
- Components of Tor
  - Tor network: A network of many **Tor relays** (proxies) for forwarding packets
  - Directory server: Lists all Tor relays and their public keys
  - Tor Browser: A web browser configured to connect to the Tor network (based on Firefox)
  - Tor onion services: Servers that can only be reached through the Tor network
  - Tor bridges: Tor relays that try to hide the fact that a user is connecting to the Tor network

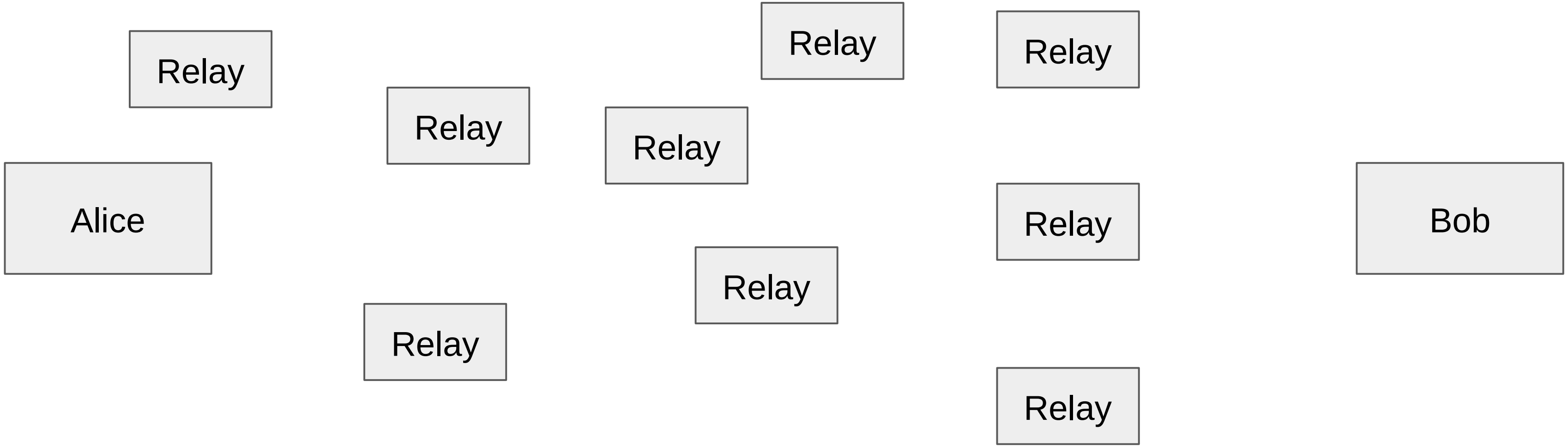
Tor Threat Model

- Security: Client anonymity and censorship resistance
  - Optional: Server anonymity with onion services
- Performance: Low latency (communication should be fast)
- Tor preserves anonymity against local adversaries
  - Example: An on-path attacker sees Alice send a message to a Tor relay, but not the final destination of the message
  - Example: The server should not know the identity of the client

Tor Circuits

- To communicate anonymously with a server, the Tor client forms a **circuit** consisting of 3 relays (by default)
    - Step 1: Query the directory server for a list of relays
    - Step 2: Choose 3 relays to form a Tor circuit
    - Step 3: Connect to the first relay, forming an end-to-end TLS connection
    - Step 4: Connect to the second relay *through* the first relay, forming an end-to-end TLS connection
    - Step 5: Connect to the third relay *through* the second relay, forming an end-to-end TLS connection
    - Step 6: Connect to the web server using HTTPS (so an end-to-end TLS connection is formed through the third relay)

Tor Circuits



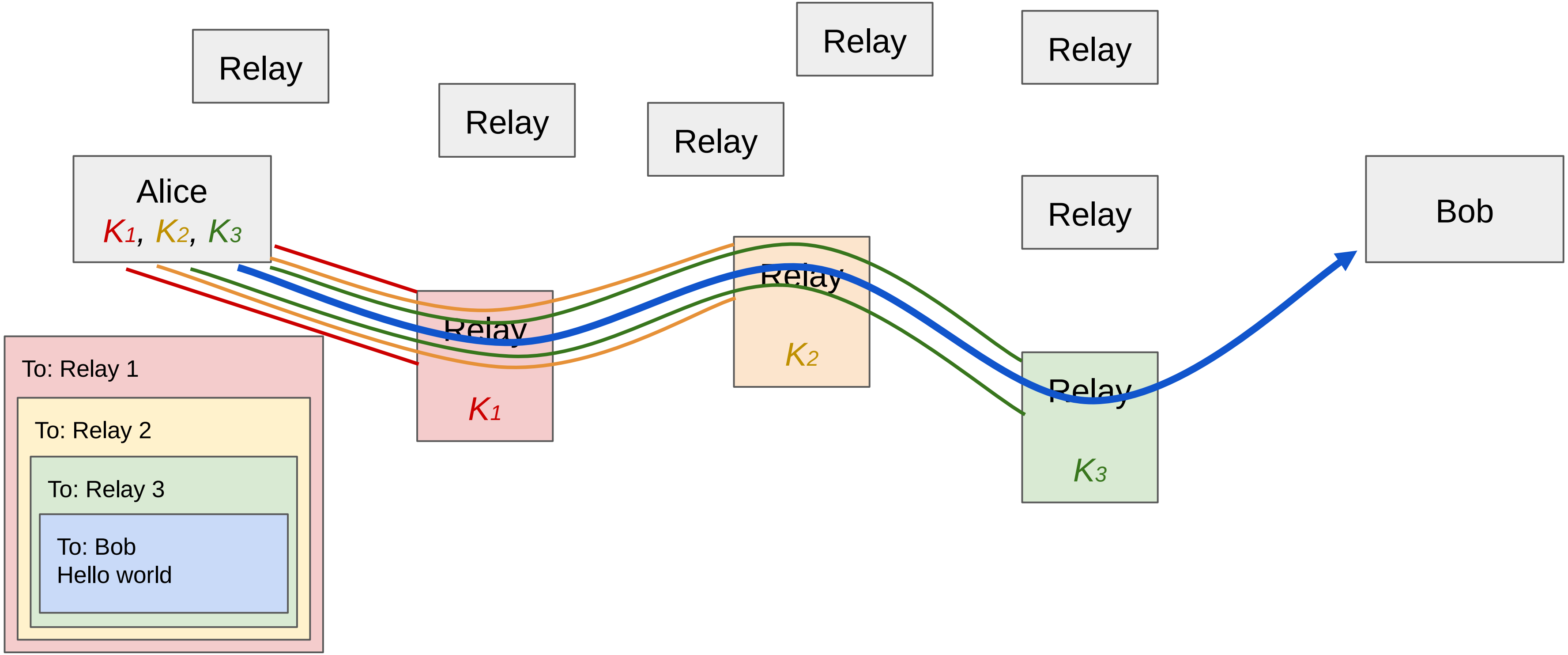
Suppose Alice wants to talk to Bob anonymously.

Alice queries the directory server and chooses 3 relays

What a Relay Does

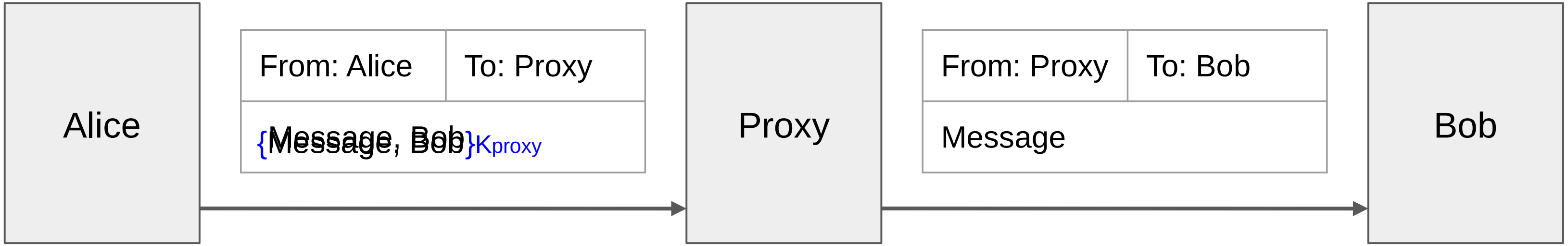
- Listen for someone to initiate a TLS connection
- When receiving a packet, decrypt using the key obtained through TLS
- Forward the packet to its destination (and forget who sent it to you)

Tor Circuits



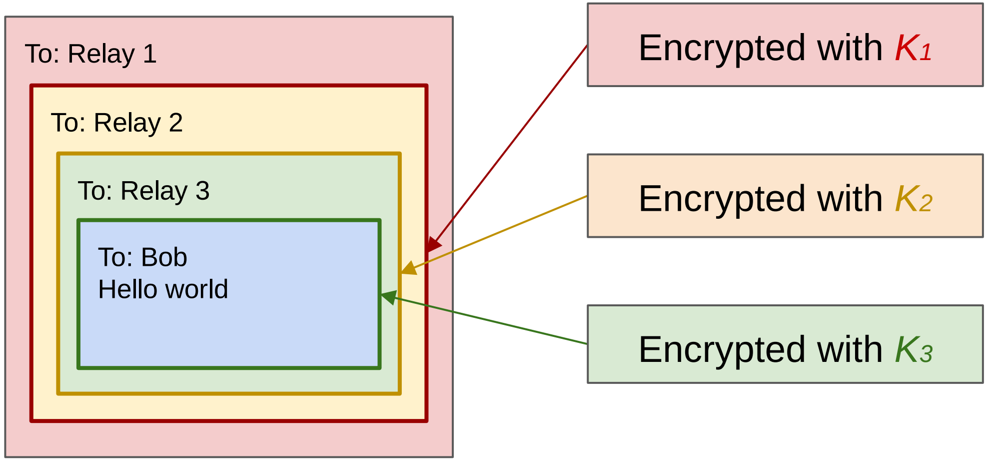
Recall: Proxy Message Encryption

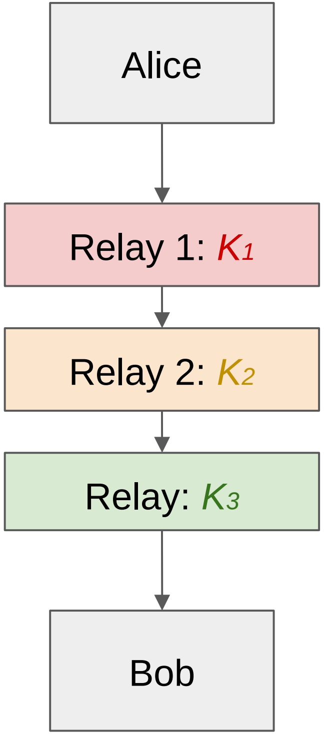
- Alice wants to send a message to Bob
- She **encrypts** the recipient's name (and message) so an eavesdropper does not see a packet with both Alice and Bob's identities in plaintext



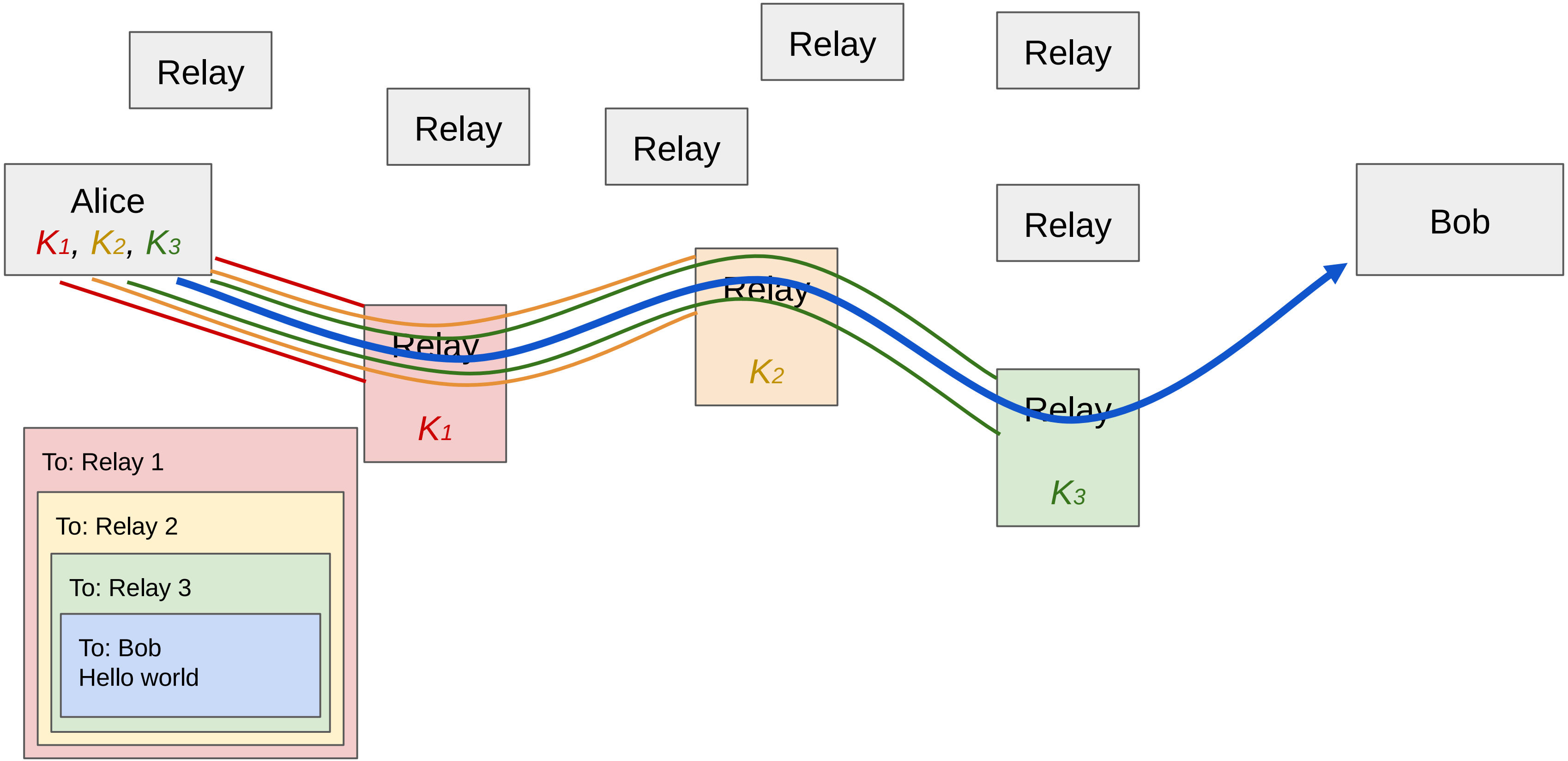
Tor Packet Construction

- Wrap the packets via encryption
  - For example, the packet sent to Bob is encrypted under K_3 since Relay 3 is the one to forward that information to Bob
- Ensures that no one can read or tamper with the messages, since these are all sent over TLS connections



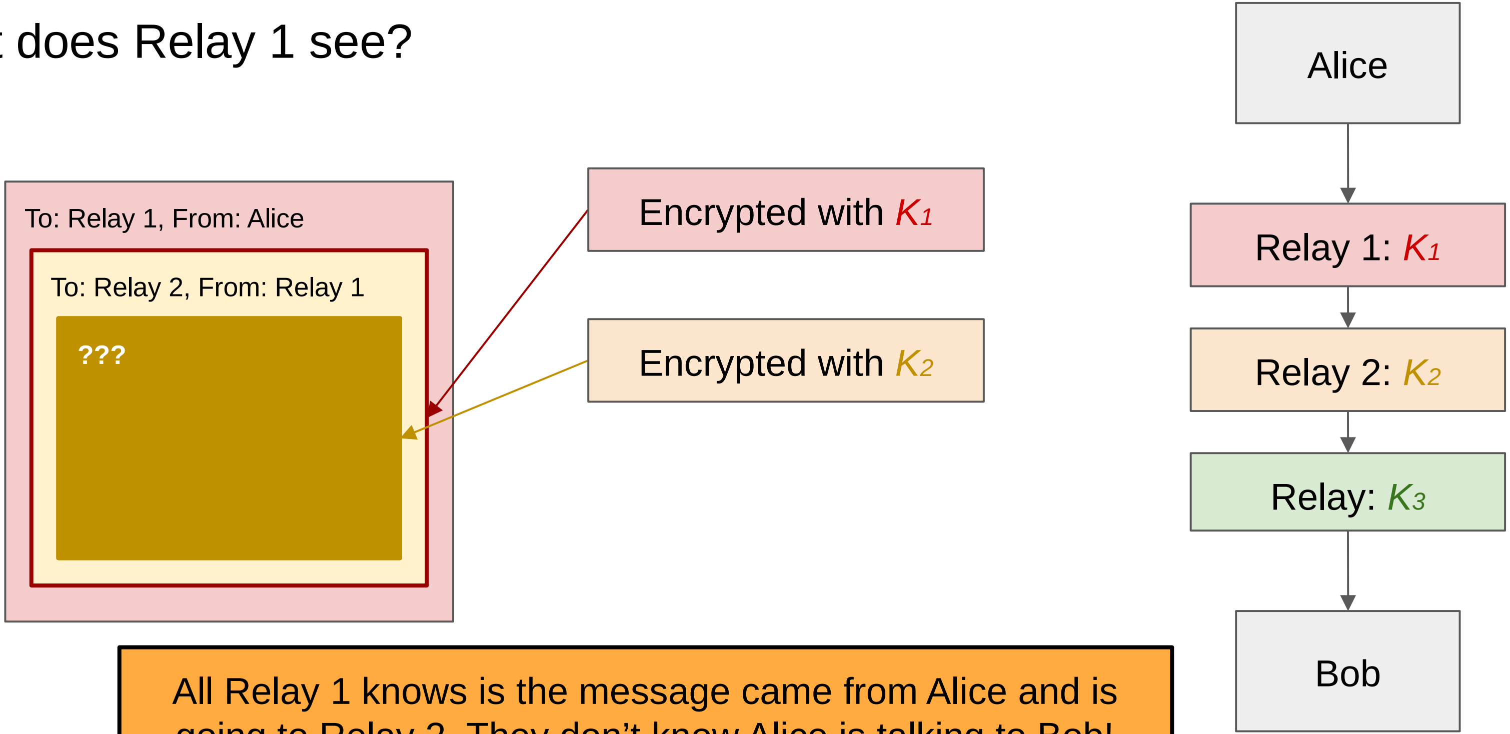


Tor Circuits



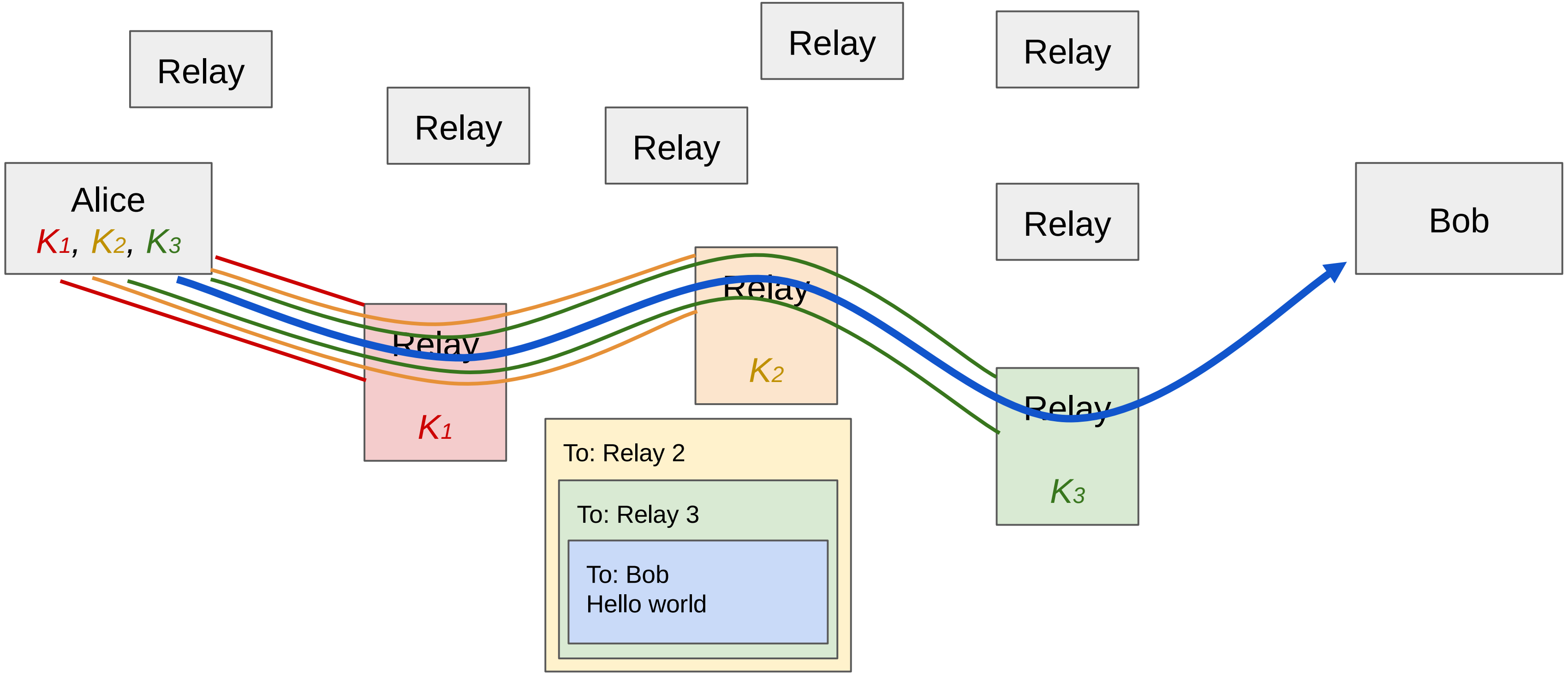
Tor Packet Construction

- What does Relay 1 see?



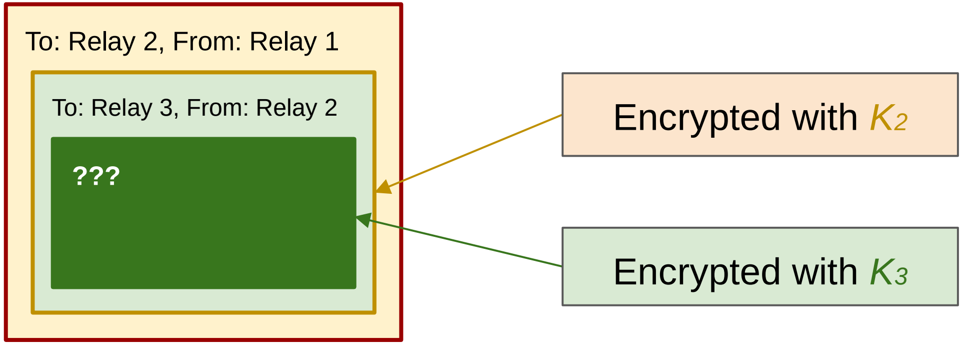
All Relay 1 knows is the message came from Alice and is going to Relay 2. They don't know Alice is talking to Bob!

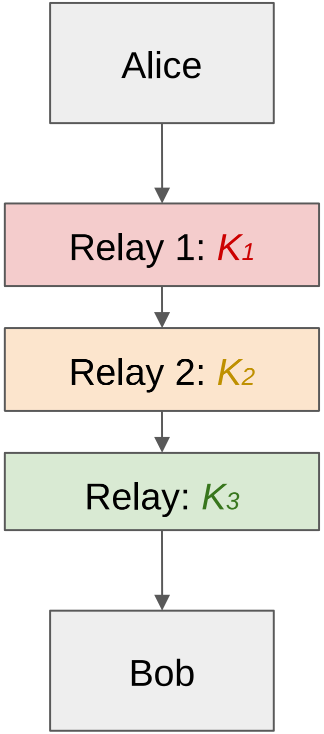
Tor Circuits



Tor Packet Construction

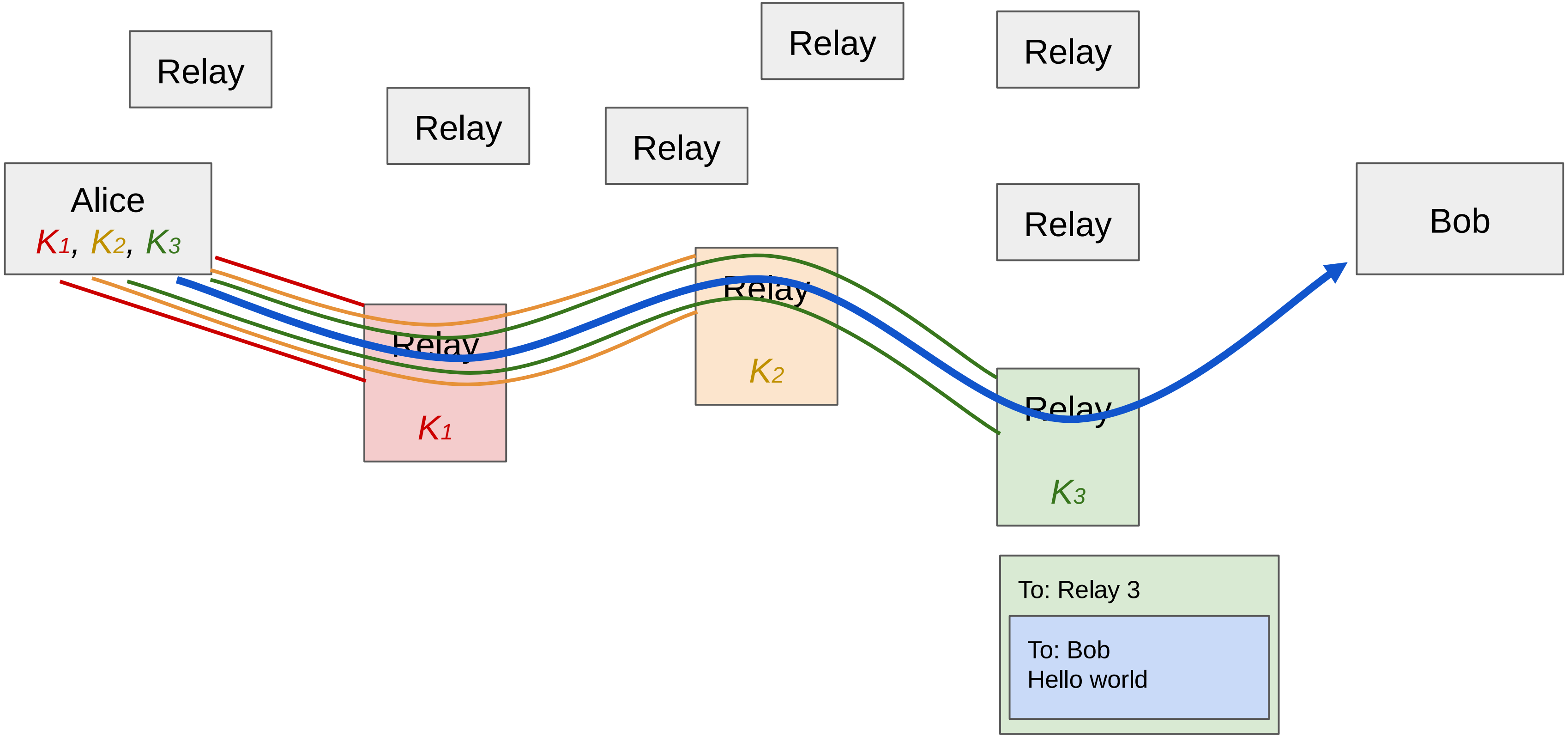
- What does Relay 2 see?





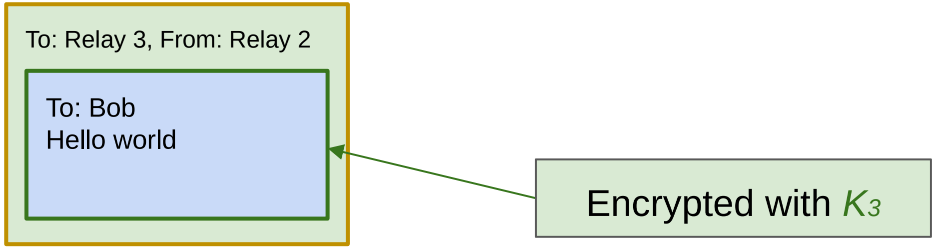
All Relay 2 knows is the message came from Relay 1 and is going to Relay 3. They know nothing about Alice and Bob!

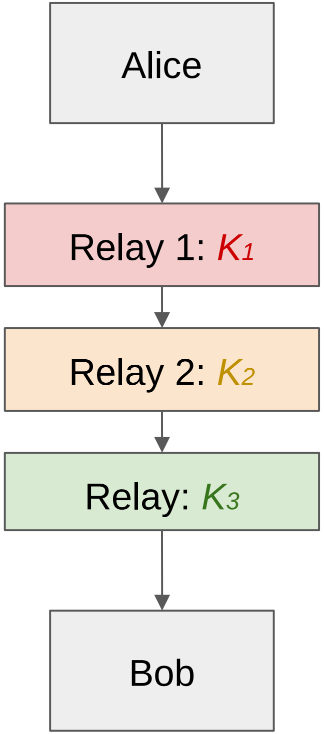
Tor Circuits



Tor Packet Construction

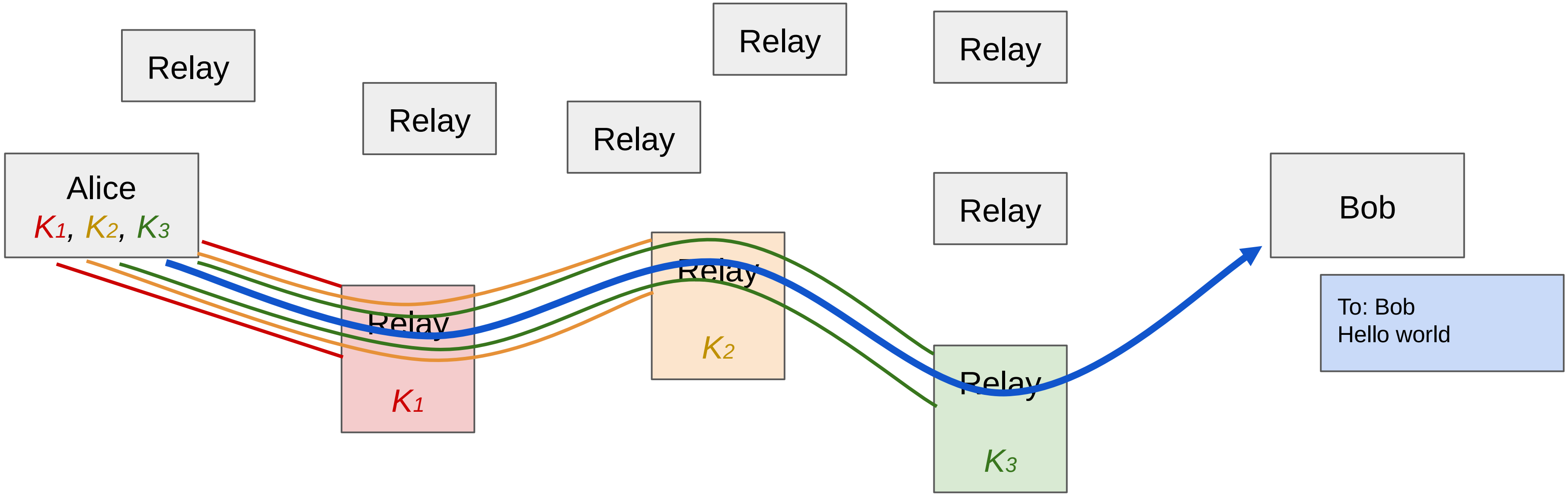
- What does Relay 3 see?



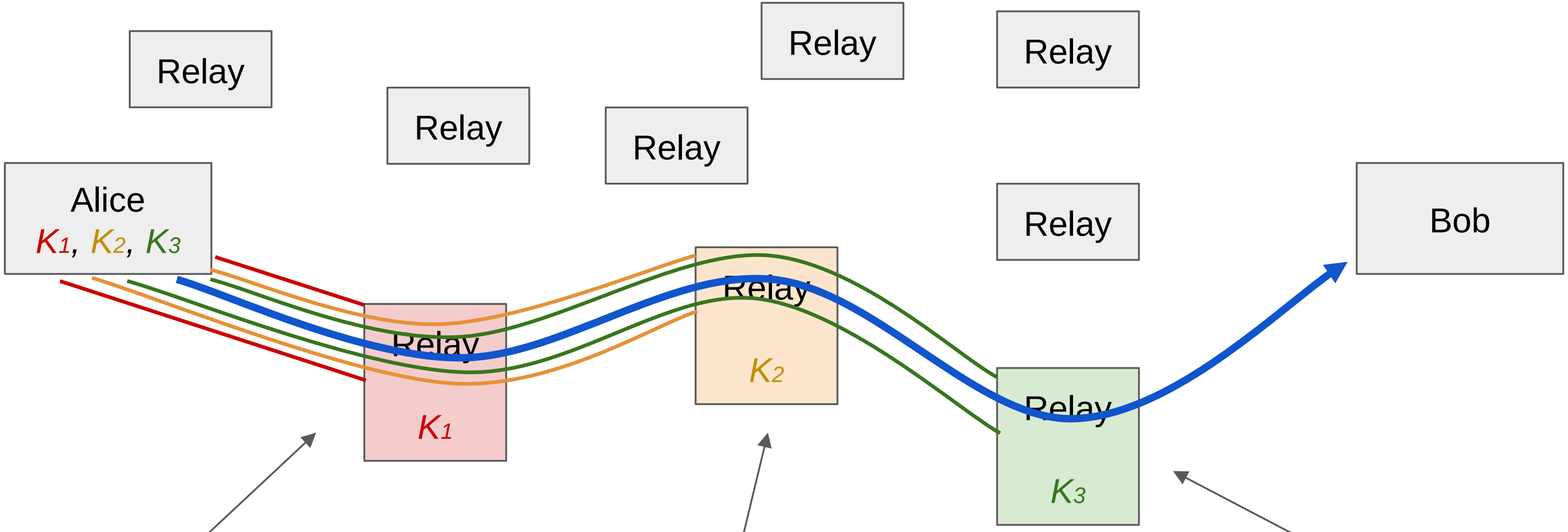


All Relay 3 knows is the message came from Relay 2 and is going to Bob. They don't know Alice sent the message!

Tor Circuits



Tor Circuits



Relay 1 knows that Alice is using Tor but *not* who Alice is talking to

Relay 2 knows nothing (other than that someone is using Tor)

Relay 3 knows that someone is talking to Bob, but not who

Tor Exit Nodes

- The exit node can see the message and the recipient (but not the sender)
- The exit node is a man-in-the-middle attacker
  - If the user is not using TLS to connect to the end host (using HTTP), the exit node can see and modify the traffic
  - If the user is using TLS (using HTTPS), the exit node cannot see or tamper with the traffic

Tor Exit Nodes in Practice

- Administrators of Tor exit nodes often receive abuse complaints
  - Users complain to the exit node
  - Users complain to the Internet service provider (ISP), which complains to the exit node
- As a result, most Tor relays choose to only be entry or intermediate nodes, not exit nodes
  - Exit node bandwidth is the bottleneck in Tor, not internal bandwidth

Tor Weaknesses: Timing Attacks

- A network attacker who has a full (**global**) view of the network can learn that Alice and Bob are talking
    - Timing attack: Observe when Alice sends a message, when Bob receives a message, and link the two together
- Global adversaries are *outside* of Tor’s threat model and are not defended against
    - Tor only defends against local adversaries with partial views of the network
    - Timing attacks could be defended against by delaying the timing of packets, but this violates Tor’s performance goal

Tor Weaknesses: Collusion

- **Collusion**: Multiple nodes working together and sharing information
  - Collusion is adversarial (dishonest) behavior
  - Honest nodes should never share information with other proxies
  - If *all* nodes in the circuit collude, anonymity is broken
  - If *at least one* node in the circuit is honest, anonymity is preserved
- The more nodes we use, the more confident we are that they are not all colluding
  - It’s much harder for 10 nodes to collude than for 2 nodes to collude
  - 3 nodes is generally considered good enough for industrial-grade security and is the default

Tor Weaknesses: Collusion

- Defense: **Guard nodes**
  - Guard nodes must have a high reputation and must have existed for a long time
  - Clients will always use a guard node as the entry node (by default) and the same guard node is used for a long period of time
    - Attackers’ nodes are unlikely to become guard nodes
    - Because clients use the same guard nodes for a long period of time, there is only a low chance that the client will switch to an attacker’s guard node

Tor Weaknesses: Distinguishable Traffic

- Tor does *not* hide the fact that you are using Tor
    - Example: A local adversary can see that you are sending packets to a Tor relay
- Anonymity only works in a crowd
    - Example: A Harvard student sends an anonymous bomb threat using Tor. The administrators noticed that only one student on the Harvard network used Tor at that time!

Tor in Practice

Tor Tradeoffs

- Benefit: Free to use
  - Tor is mostly funded by the US government
  - Users “pay” by providing traffic for other users to hide in (recall: you don’t want to be the only user on the network using Tor)
- Drawback: Performance
  - Latency is significantly worse: Packets need to make more hops across the network
- Drawback: Full anonymity requires usability tradeoffs
  - All Tor browsers need the exact same configuration, so they don’t save your history
  - They even recommend keeping the browser window size constant, which can be annoying!

Hosting Illegal Services on Tor

- Tor onion services are often used for services widely considered illegal around the world
  - Legitimate hosting services like Cloudflare will refuse to host these services
  - Most countries will take legal action against these services if hosted on the regular web
- **Dark markets**: Marketplaces for buying and selling illegal goods
  - Transactions processed with a censorship-resistant currency like Bitcoin
    - Services like PayPal will refuse to process illegal transactions
  - Can only be accessed as a Tor onion service
- **Cybercrime forums**: Websites for discussing illegal activity

Tor Onion Services

- Websites or services hosted inside the Tor network.
  - Sometimes called the **dark web**
- Sometimes, the server wants to be anonymous, so no one knows where the server is located
  - Standard domain names translate to IP addresses, which would break server anonymity
  - Instead, identify servers using the hash of the service's public key encoded as a .onion address
  - Example: https://www.bbcnewsd73hkzno2ini43t4gblxvycyac5aw4gnv7t2rccijh7745uqd.onion/
  - Not accessible via normal browsers.
- Both **client** and **server** stay anonymous.
- Details if you are interested: https://community.torproject.org/onion-services/overview/

Summary: Tor

- Anonymity (concealing one's identity) can be difficult to achieve on the web
- Proxies and VPNs relay traffic through a single machine to conceal your identity from the end server
  - Issue: The single relay knows who you are and what you are doing, which is not anonymous!
- Tor routes your traffic through multiple machines
  - No one machine knows both who you are and what you are doing
  - Circuits are established by performing TLS handshakes with three nodes, nesting encrypted channels
  - Weakness: Timing attacks allow global adversaries to see when packets exit and leave the Tor network, deanonymizing users
  - Weakness: Collusion between nodes can deanonymize users by working together
  - Weakness: Tor traffic is distinguishable from normal traffic, allowing it to be censored and blocked
    - Defense: Bridges and pluggable transports

Summary: Tor

- Onion services provide anonymity for the server, in addition to the client
- Tor in practice
  - Often used to evade censorship -- Tor and censors are in an arms race
  - Illegal services often use Tor because it conceals their identity from authorities